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Endometriosis: Living as the Overlooked 10%

Discussing the workings of endometriosis and the system that refuses to

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Abstract

Endometriosis is a chronic inflammatory disorder affecting up to 10% of women worldwide, which is heavily under researched. This condition consists of tissue resembling the cells lining the uterus migrating to other parts of the body, forming lesions and scarring much like cancer. These lesions inflame, and can cause a myriad of symptoms, all depending on the patient. This condition can affect any area of life, from fertility to mental health, and hasn't received the attention it deserves despite its potentially detrimental impacts. For decades, the medical community has dismissed the disorder and overlooked women's health. In this article, not only are the physical symptoms discussed, but the barriers women with endometriosis face today, the mental toll it takes, and what exactly has stopped us from finding a cure.

Introduction

If you follow any biomedical engineers or healthcare professionals on social media, you have no doubt seen outrage expressed towards the lack of resources going towards women's health and research, particularly chronic conditions. One of these conditions is endometriosis, a chronic inflammatory ailment, caused by the migration of cells resembling endometrial tissue outside of the uterus and leading to detrimental and multilateral side effects. From sexual function to mental health, endometriosis has a negative effect on many domains of life which leaves many feeling powerless and at the mercy of their disorder. This condition affects up to 10% of women worldwide, often going undiagnosed or mistreated due to systemic barriers and biases in healthcare (Chauhan et al., 2022).

Endometriosis is not just a phenomenon seen in medical textbooks: it causes intense pain and suffering for many women worldwide. Nevertheless, this disorder has been consistently dismissed by medical researchers and denied grants. It eventually leads you to ask one question: 'Why is this happening?' The main explanation is that the medical system today still holds the ideals of yesterday, forcing marginalized groups to pay the price for medical negligence. The only way to combat the real-life barriers in healthcare we see today, is to take action. Whether it be raising awareness, or learning more about the condition to advocate for yourself, you can make a difference.

Signs and Symptoms

Endometriosis is unique in the symptoms it causes, both due to the disorder's variability and of its unclear causes. A symptom that is common for many women living with endometriosis is dysmenorrhea, or painful periods (Chauhan et al., 2022). When menstruation causes a high level of pain, it can usually be attributed to an overproduction of hormones, leading to lower back pain, heavy bleeding, longer menstrual phases and pelvic pain. Another key symptom is dyspareunia, which is pain during sex. There are two main types of dyspareunia, deep and superficial. Superficial dyspareunia is pain limited to the vaginal opening and the vulva. Deep dyspareunia, however, causes pain to radiate to the lower pelvis (Chauhan et al., 2022). Other symptoms like infertility, chronic pain, impaired sexual function, internal bleeding and impaired function of certain systems (i.e. metabolic) are also integral parts of the disease.

This list of symptoms doesn't apply to everyone, in fact, many women experience different symptoms based on the location of their endometrial lesions. For example, if a lesion is located in your digestive tract, you may have bloody stool during your menstrual cycle (Yin et al., 2023). This is due to the endometrial cells in your gut shedding its lining, just like during your period. Nevertheless, patients may also be asymptomatic, or only display a limited amount of symptoms. Combined with the lack of knowledge concerning causes for symptoms, and the migration of tissue into many parts of the body, endometriosis is difficult for medical professionals to spot without imaging, analyzing saliva or invasive surgery with the training and technology at their disposal today.

Note: Symptoms vary widely between individuals; readers should not self-diagnose based on this list. If someone suspects they have this condition, they should consult a healthcare provider.

Causes and Risk Factors

Despite the overlooking of endometriosis in research, scientists have managed to narrow down causes and risk factors for the disorder in recent years. Many recent studies have successfully hypothesized some probable causes for endometriosis, with the main identified cause being something called retrograde menstruation (Chauhan et al., 2022). Retrograde menstruation consists of cells from the uterine lining travelling up the fallopian tubes during the menstrual phase. Due to the fallopian tube not being physically attached to the ovaries, the tissue migrates to other parts of the abdominal cavity. This phenomenon also plays a role in ectopic pregnancies, meaning pregnancies where the fertilized egg migrates and grows in other parts of the abdominal cavity. Retrograde menstruation can happen to almost anyone with a cycle, and already happens to 76-90% of women with patent fallopian tubes, a term used to designate healthy tubes that allow ovum to travel through (Chauhan et al., 2022).

In recent years, researchers have succeeded in identifying risk factors with both loose and strong correlations. Firstly, there is an element of genetic predisposition. Recent research has shown that endometriosis tends to be observed in families, and therefore inherited (Hansen & Eyster, 2010). Moreover, certain variables such as diet have been loosely correlated with endometriosis, demonstrating the potential link between trans fat and red meat consumption (Arab et al., 2022). Notwithstanding some research being done on this theorized link, it remains unproved as little research has been done. Furthermore, previous surgery in the lower abdominal

area can increase the likelihood of endometrial cells spreading, causing something called 'scar endometriosis'. For example, patients may be at risk after cesarean sections due to the possibility of tissue breaking away from the main incision and migrating to other parts of the body (Chauhan et al., 2022). The large advancements in our knowledge about endometriosis is absolutely a cause for celebration. However, due to the newness of most recent research, certain domains have been underexplored and there is still much work to be done to fully understand this complex gynecological pathology.

Diagnosis

In the past, the only way women with endometriosis could obtain a diagnosis was through invasive procedures like exploratory surgery, or large surgeries where lesions were spontaneously discovered. Today, many advancements have been made in diagnosing patients with endometriosis that are more cost effective, less invasive and more efficient.

First and foremost, imaging is the most commonly used procedure to detect endometriosis. MRIs, ultrasounds, and other forms of medical imaging help greatly in identifying the locations and number of lesions, allowing for more targeted treatment and a personalized plan moving forward (Mayo Clinic, 2024). Additionally, laparoscopy is another commonly used way to determine whether a patient has endometriosis. Through minimally invasive surgery assisted with a flexible camera, surgeons can perform operations in the abdominal and pelvic region with ease (Mayo Clinic, 2024). This allows for lesions to be removed immediately, and is done after imaging has been performed to either eliminate the excess endometrial tissue or to further explore the cavity. Finally, new research has discovered that endometriosis can be detected through saliva analysis, looking for a microRNA (or miRNA) marker that indicates the presence of endometrial lesions (Nigdelis et al., 2024). Despite this treatment being revolutionary, it is still new and needs to be further explored before widespread implementation (Nigdelis et al., 2024). This does not negate the huge importance of more minimally invasive avenues of diagnosis, particularly for adolescents or individuals who would be at risk or uncomfortable during invasive procedures.

Finding avenues of detecting endometriosis medically isn't the greatest barrier for those possessing the disorder; it's the huge gap in time between experiencing symptoms and formal diagnosis. For many women, this means months of being referred between specialists just to be told the same thing from every one.

Combined with the time it takes naturally for symptoms to be concerning, this wait time becomes painful and insufferable. Instead of walking into an appointment without knowing what you need to do to be screened, seek to learn more about what's to come and what screenings you need to exhaust the possibility of endometriosis. There are many services which help clients to prepare questions for appointments, for example when a certain disease is to be the suspected cause of your symptoms. Advocating for yourself and your health is one of the most significant factors contributing to getting you the diagnosis to access the care you need, especially in a system where women's reproductive health has only recently been made a priority.

Treatment

As of now, there has not been a discovered cure for endometriosis. But, the disorder is highly treatable through various methods such as surgery and hormonal treatments. These treatments often range depending on the stage of endometriosis, and where lesions are located. Like cancer, there are four stages of endometriosis, with Stage IV being considered the most severe (Chauhan et al., 2022), depending on the amount of lesions and adhesions present. Stage I consists of small lesions (active tissue that is responsive to hormones) with no lesions (scar tissue that has become unresponsive to hormones). Stage II consists of lesions covering the surface of the ovaries, and some very mild adhesions, whilst Stage III entails the deepening of lesions on the ovaries and the potential spread of adhesions to the fallopian tubes and ovaries. Stage IV is the most severe stage, with dense adhesions and lesions all over the ovaries, fallopian tubes and endometriomas in the ovaries. Endometriomas are large ovarian cysts, filled with menstrual blood accumulated over time (Chauhan et al., 2022). It is important to note that a specific stage doesn't necessarily align with a set of symptoms, it only specifies the progression and appearance of the lesions, adhesions and endometrioma. These Stages are the basis for treatment, along with the patient's goals pertaining to fertility.

For most cases of endometriosis, medical professionals opt for drug-based treatment plans before moving on to surgery if the case progresses (Mayo Clinic, 2024). The main types of medicine administered to patients with endometriosis are painkillers, hormone-based treatments and fertility treatments. All of the following medications and treatments aim to balance or decrease the amount of hormones in your bloodstream, which tricks the lesions in the body into minimizing how much they inflame during flare-ups. Some aim to eliminate the menstrual cycle by simulating menopause, whereas others may also possess the ability to shrink endometrial

lesions. Many treatments can also be administered at once, depending on dosage, the progression of the illness and what works best for each individual. If hormonal treatment fails, only then will your doctor suggest surgery to remove the lesions and/or endometrioma (Chauhan et al., 2022).

The main options that a patient will be provided will most likely be laparoscopy or hysterectomy, depending on progression (Stage I, II, III or IV) and the fertility goals of the patient. Laparoscopy, explained in a previous section, is a minimally invasive, camera assisted surgery where the abdominal cavity is lightly inflated and long tools are used to explore the site. This form of surgery aims to conserve the structures underneath it, and is usually offered to patients with mid stage endometriosis or patients who want children in the future. If removing the lesions and endometrioma laparoscopically is not possible, or the adhesions are too deep for a good turnout, the patient is usually offered the choice to attempt laparoscopic surgery or to undergo a hysterectomy. There are multiple types of hysterectomies depending on the size of the affected area, including an oophorectomy which is when the ovaries are completely removed (Mayo Clinic, 2024). Depending on how the disorder has spread and your desired outcome, your medical professional will recommend the procedure that's best for you. Remember that your needs come first, and that they should be reflected in the care that you are provided. If you are not given the care you request, do not be afraid to advocate for yourself. Whether it's bringing in someone else to your appointments to help push for opportunities you want to explore, or receiving a referral to another specialist, do what you must to receive the guidance, respect and care you deserve.

Emphasize: Treatment plans are highly individualized; readers should always consult their personal physician before starting or modifying treatment.

Living with Endometriosis

Most of this text has discussed the overall symptoms, risk factors and physical tolls that endometriosis takes on those who have it, but living with this disorder is a different topic entirely. Symptoms have a much broader effect than just physical symptoms, they have a large impact on the daily life and the mental health of those who possess them. It deeply affects overall quality of life, social fulfillment and interactions with others. It may even negatively affect individuals financially, as endometriosis can make it difficult to even get out of bed. It can even complicate romantic relationships, due to impaired sexual function caused by the condition. In a study assessing the overall mental health of women with endometriosis before and after treatment, it was found that before treatment, "scores of patients with

endometriosis were significantly diminished in comparison to controls, especially in general health, vitality, mental health, and emotional role functioning domains" (Rempert et al., 2024). Moreover, 43.1% of women with endometriosis suffer from moderate to severe depression (Rempert et al., 2024). This indicates an obvious disparity between those with and without endometriosis, and the profound effect chronic pain disorders can have on all areas of life. Luckily, after treatment, the same study saw that "surgical, pharmaceutical (estrogen-progestin), or combined intervention led to a significant increase in overall QoL scoring ($p < 0.03$), particularly in the domains of bodily pain, general health, and social functioning" (Rempert et al., 2024).

You must be thinking that this is amazing, and that women only have to get treatment to see an improvement in mental health, especially after those promising results. However, the reality of this diagnosis isn't as straightforward as you may think. Due to biases and limited research into the disorder, diagnoses are extremely difficult to receive, and requires extensive appointments with medical professionals. For many women, this is difficult to accomplish because of extensive referrals and long wait times in the public system. Combine this with infrastructure that has historically denied and dismissed the pain of women, and obtaining a diagnosis seems completely out of reach for most women. And for this reason, many women cannot access the care they need to help alleviate their debilitating symptoms.

Conclusion

After reading the bulk of this article, try imagining what life could be like for the 5-10% of women that have this disease. The chronic pain, the heavy periods, the crippling symptoms and the inaccessibility of a diagnosis. Working, eating and even sleeping are affected, with many endometriosis patients having negatively impacted sleep quality (Rempert et al., 2024). Now, let's come back to reality, where you're back in your own body. Whether you're part of that 5-10% or not, there is so much you can do to help those who are. You might not have endometriosis, but my hope is that you understand how it can take over the lives of millions without warning. Talking about it, supporting those who have it and advocating for better care can change their lives for the better. Advocacy is the catalyst for positive change, and that can only happen by joining and amplifying the discourse that exists today and one day, you might be part of the reason that something happens to help that 5-10% of women. Because in the end, endometriosis cannot simply be reduced to a pathology, because it has so much more gravity to those who live through it every day.

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